Do flood susceptibility models know more than the flood record? Evidence from radar-observed floods in Bangladesh

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Running title: Flood records and susceptibility models

Abstract

Flood susceptibility maps are judged by how well they rank the ground that floods, yet unlike forecasts, which are scored against persistence, they are rarely compared with the simplest alternative, the record of where floods have already been. This study makes that comparison for infrastructure mapped in OpenStreetMap in Bangladesh, with flood extents mapped from Sentinel-1 radar as labels and reference and every model scored on 10 km blocks held out from training. Each flood year was predicted from the years before it, giving 15 tests in three study regions and four independent validation regions. The record's lead over a gradient-boosted terrain model fell as the share of a flood reaching ground no earlier flood had reached rose (Spearman $\rho = -0.775$, $p < 0.001$), whatever the length of the record, and the two were level when about 33.4 % of flooded assets stood on such new ground. In Rangpur and Rajshahi, where only 6.8 % of the 2024 river flooding reached new ground, the record ranked it at 0.896 against 0.801 for the model and kept its lead with masks re-mapped from a single satellite, with other training targets and with 40 km blocks, while on the cyclone-struck coast, where most flooding reached new ground, the model led. A global hydrodynamic hazard map was no stronger a rival than the model, and the model given the record as a feature beat both of its parts in 5 of 8 tests, all riverine. Radar-observed extents also made better training labels than terrain thresholds, by 0.101, 0.087 and 0.080 in AUC on held-out areas. The radar masks agreed only moderately with an independent MODIS map, so every score measures agreement with flooding as radar sees it. Susceptibility studies with a radar flood history should report that record as a baseline and score their models by the share of its ranking error they remove.

Keywords: flood susceptibility; Sentinel-1; spatial cross-validation; temporal validation; gradient boosting; graph neural network; persistence baseline; Bangladesh

1 Introduction

Riverine and flash floods recur across northern Bangladesh in most monsoon seasons and cyclone surges strike its south-west coast, yet the susceptibility maps meant to show where flooding would do most harm are rarely tested against floods that actually occurred. The studies closest to this one report best-model AUCs between 0.87 and 0.98 (Islam et al., 2021; Talukdar et al., 2020; Islam and Chowdhury, 2024; Adnan et al., 2023), and two choices decide what such a figure means. The first is where the flood locations come from, usually a few hundred to a few thousand points drawn from field surveys, local knowledge or satellite flood maps. The second is how the test set is separated from the training set, since a random split of spatially clustered points tests a model mostly beside its own training points, a design known to inflate accuracy (Roberts et al., 2017; Ploton et al., 2020).

Both choices can now be made differently, because Sentinel-1 radar sees through monsoon cloud and has been processed into flood maps with overall accuracies above 94 % (Twele et al., 2016; Uddin et al., 2019). Every pixel observed to flood can therefore serve as a training label (Islam and Chowdhury, 2024), and whole areas can be held out so that the test asks how well a model carries to ground it has not seen. A radar record of several floods also offers the reference that forecasting takes for granted, since a weather or river forecast is judged by the skill it adds over persistence or climatology (Murphy, 1988; Pappenberger et al., 2015). For a susceptibility map that reference is the map of where water has already been, which susceptibility studies rarely report.

This study asks two questions of flood susceptibility models for mapped infrastructure in three regions with contrasting regimes, riverine flooding in Rangpur and Rajshahi, flash flooding in the Sylhet haor basin and cyclone surge on the south-west coast. The first, and the most consequential for how published maps should be read, is whether a model anticipates a later flood better than the record of earlier ones, and on what the answer depends. It is tested on every flood year that has an earlier one, in the study regions and in four validation regions that share no asset with them, against a global hydrodynamic hazard map as well as the record. The second is whether radar-observed flood extents make better training labels than terrain thresholds for ground a model has not seen. The maps, data and code are released openly.

2 Related work

2.1 Flood susceptibility in Bangladesh

Table 1 summarises the studies closest to this one, each checked against its full text. They report high accuracies, from 0.81–0.87 for five models in the Teesta basin (Islam et al., 2021) to 0.984 for a random forest on the south-west coast (Adnan et al., 2023), but every study whose split could be read divided its points at random except one, which held out five regions within the 2024 Feni flash flood and found that random splits raised AUC by 0.06–0.09 (Diptho et al., 2026). The inventories also differ in kind, from points identified by field survey and local residents (Islam et al., 2021; Talukdar et al., 2020) or a water-board survey (Rudra and Sarkar, 2023) to pixels sampled from satellite flood maps (Islam and Chowdhury, 2024; Adnan et al., 2023; Diptho et al., 2026), so the same accuracy figure can measure agreement with quite different things. Two studies compared the maps their models produced, and those comparisons are the more instructive. On the south-west coast four models each classified roughly 60 % of the land as flood-prone, yet their pixel-wise correlations ranged from 0.62 to 0.91 (Adnan et al., 2023). In one Sylhet sub-district a random forest and a support vector machine with AUCs of 0.964 and 0.954 agreed on only 73.3 % of the map (Islam and Chowdhury, 2024). Models that score equally well can therefore disagree about where flooding will happen, which is the property a map exists to convey. A multi-criteria hazard map of the Sangu basin placed 39 % of roads and 25 % of education buildings in its highest hazard class (Zzaman et al., 2021), which shows why infrastructure exposure matters to planners.

2.2 Labels and validation

Radar offers training labels that cover every pixel, and Sentinel-1 flood maps have reached overall accuracies of 94.0–96.1 % (Twele et al., 2016) and 96.44 % for the 2017 Bangladesh floods (Uddin et al., 2019). Daily MODIS imagery has mapped 913 large floods worldwide (Tellman et al., 2021), although radar still struggles in vegetated and urban areas (Amitrano et al., 2024), and Sentinel-1 flood maps have already been used to estimate flood exposure across a whole country (Sherpa and Shirzaei, 2022). One Bangladesh study has already built its inventory from Sentinel-1, for a single Sylhet flood and with a random split (Islam and Chowdhury, 2024). Where no inventory exists, labels are often thresholded from the topographic wetness index (Beven and Kirkby, 1979) and height above nearest drainage (Nobre et al., 2011), the same indices that serve as predictors, so a model scored against such labels can succeed by relearning the rule that made them.

Random cross-validation of spatially structured data underestimates prediction error, and splitting by area can address this (Roberts et al., 2017). Splitting by time, training on past events and testing on later ones, has long been standard for landslides (Chung and Fabbri, 2003; Guzzetti et al., 2006), but among the Bangladesh flood studies only one map has been tested on later floods, a multi-criteria map of Sylhet scored against the floods of June 2022 and June 2024 (Swarup and Nury, 2026), and none has been set against the record of earlier floods. Where records and models have been compared, it was as hazard maps, with satellite records showing less frequent flooding than global flood models predict (Hawker et al., 2020). Such records already exist for Bangladesh, as a national Sentinel-1 flood-frequency map (Haghighi, 2022), a weekly inundation series built for flood insurance (Thomas et al., 2023) and a two-decade reconstruction (Giezendanner et al., 2023). Blocking can force a model to extrapolate (Roberts et al., 2017) and has no theoretical basis for estimating map accuracy (Wadoux et al., 2021), so blocked scores are read here as performance on unseen areas of similar terrain, not as design-based map accuracy. Calibration is treated apart from ranking, by isotonic regression (Zadrozny and Elkan, 2002; Niculescu-Mizil and Caruana, 2005), and among the Bangladesh studies only Diptho et al. (2026) calibrated its scores, within a single flood event.

2.3 Past events as the reference

Other hazard fields score their models against the record of past events as a matter of course. In earthquake forecasting the reference is relative intensity, a map built only from smoothed past seismicity, and in a ten-year experiment in California neither a pattern-recognition model nor the United States national seismic hazard map gained significantly over it (Zechar and Jordan, 2008). That test scores each forecast by its Molchan trajectory, the share of later events missed against the share of space flagged, summarised as an area skill score. Crime analysts likewise build hotspot maps from earlier crimes and judge them by the share of later crimes they hold over the share of area they flag (Chainey et al., 2008). For landslides, models trained on a single extreme year failed to hindcast other years while inventories of more than six to eight years hindcast well (Jones et al., 2021), so a test on one later year says little about the next. The skill a forecast earns also depends on the benchmark it is set against, and the most informative benchmark is the one hardest to beat (Pappenberger et al., 2015). For a map of where floods will occur that benchmark is the map of where they already have, and no flood susceptibility study found here has set its model against it.

3 Study area and data

Three regions with contrasting flood regimes are studied, riverine flooding in Rangpur and Rajshahi, flash flooding in the Sylhet haor basin and cyclone surge on the south-west coast (Fig. 1a, Table 2), and they share one set of model settings so that results stay comparable. Rangpur and Rajshahi, in the north-west, lie on the floodplains of the Teesta and the Brahmaputra–Jamuna, where monsoon rivers rise above their danger levels and stay there for days to weeks, and all five mapped floods fell between late June and October (Table 4). The Sylhet region holds the haor basin, a low bowl of seasonal wetlands fed by rivers from the hills to the north and east, where flash floods rise quickly and can arrive before the monsoon, as in April 2017, when they struck before the rice harvest. The south-west coast is a delta of tidal channels and polders, embanked islands of farmland, where flooding follows the landfall of a cyclone and the overtopping or breach of embankments rather than the stage of a river. The regions hold 23,016, 13,374 and 29,545 mapped assets, across parts of 21, 10 and 16 districts. Four further regions serve only to validate the comparison with the flood record (Fig. 1a, e–h, Table 4): the Jamuna east bank, the central districts around

Study	Region	Model	Flood locations and split	Reported accuracy
Talukdar et al. (2020)	Teesta basin	bagging ensembles	207 flood, 206 non-flood points from inundation maps, topographic maps and local survey, random 80:20	AUC 0.945 (BgM5P)
Islam et al. (2021)	Teesta basin	Dagging, RS, RF, ANN, SVM	167 flood, 167 non-flood points from records, fieldwork and Google Earth, random 80:20	AUC 0.81–0.87 (validation)
Islam and Chowdhury (2024)	Sylhet, one sub-district	RF, SVM	1,500 flooded, 1,500 dry pixels from Sentinel-1, one event (June 2022), random 70:30	AUC 0.964 (RF)
Rudra and Sarkar (2023)	Sylhet division	ANN	1,280 flood and non-flood points from a Bangladesh Water Board survey, random 75:25	prediction rate 89–92 %
Swarup and Nury (2026)	Sylhet division	AHP, 19 indicators	independent flood inventories of June 2022 and June 2024, no training data	AUC 0.836 (2022), 0.794 (2024)
Diptho et al. (2026)	Feni basin	LightGBM, XGBoost, RF	1,510 flooded and dry pixels from the observed August 2024 inundation, five spatial blocks	spatial AUC 0.938 (LightGBM)
Adnan et al. (2023)	south-west coast	RF, KNN, MLP, GA-RBF-SVR	965 points sampled from a satellite flood-frequency map (1988–2012), 70:30 split	AUC 0.984 (RF), inter-model r 0.62–0.91
This system	Rangpur & Rajshahi	Gradient boosting	Sentinel-1, 5 events, 10 km blocks	mean AUC 0.817
This system	Sylhet	Gradient boosting	Sentinel-1, 4 events, 10 km blocks	mean AUC 0.857
This system	south-west coast	Gradient boosting	Sentinel-1, 4 cyclones, 10 km blocks	mean AUC 0.839

Table 1. Flood susceptibility studies in Bangladesh compared with this system. Every study was read in full, and where a figure is given for training and validation separately the validation figure is shown. This system's figures are means over 20 assignments of spatially held-out blocks, scored against observed floods, so the columns are not directly comparable, for the reasons given in the text.

Dhaka, the Chattogram coast and the west-central districts, where both mapped events were waterlogging after heavy rain rather than river floods. None holds an asset that a study box also holds. Every dataset the pipeline ingests is openly licensed (Table 3). OpenStreetMap coverage is uneven, so an area with few mapped assets shows low exposure whether or not much stands there, and WorldPop's constrained product carries population only where buildings were detected in imagery.

4 Methods

The pipeline runs as a chain of steps (Fig. 2), and this section follows it in order. Settings not given here are in the Supporting Information (Section S1).

4.1 Assets, terrain and features

Assets are fetched from OpenStreetMap in three tag groups, lifelines (hospitals, schools, shelters, bridges, embankments), transport (primary to trunk roads, railways) and agriculture (farmland, aquaculture, canals, markets), and each is classified into one asset type. The configured boxes cross Bangladesh's land border, so assets in India, or on land more than about 1 km outside the geoBoundaries border, are dropped, while those on chars, coastal ground and offshore are kept. Grid cells and grid-level validation are restricted the same way. From the 30 m SRTM elevation model, reprojected to UTM zone 46N, the pipeline derives slope, the topographic wetness index (TWI), height above nearest drainage (HAND) and D8 flow accumulation. FABDEM, which removes forest and building heights (Hawker et al., 2022), would suit a flat delta better, but its non-commercial, share-alike licence conflicts with the CC BY 4.0 data archive, and the vertical error of SRTM may be one reason, untested, why TWI and HAND add little once elevation is known (Section 5.6). Each asset carries these five terrain values at its representative point, WorldPop population density, its distance to the nearest hospital, school, flood shelter, road and irrigation channel, and its asset type, all standardised, and the pipeline stops if any feature is constant or the labels hold a single class.

4.2 Flood extents from Sentinel-1

Flood extents mapped from Sentinel-1 radar supply both the training labels and the reference against which the models are scored. Following the change-detection method recommended by UN-SPIDER (UN-SPIDER, 2019), an approach whose best Sentinel-1 variant reached an overall accuracy of 86.3 % against reference flood maps in an evaluation for Thailand (Dadhich et al., 2026), the minimum VV backscatter in each flood window is compared in Google Earth Engine with the median of a dry-season baseline from the same year. A pixel is flagged as flooded where it is below −16 dB and at least 3 dB darker than its baseline. Permanent water from the JRC record (Pekel et al., 2016), slopes above 5° and specks under eight pixels are removed, and the event masks are averaged into a flood-frequency layer (Fig. 1b–d). Five events are mapped in Rangpur and Rajshahi, four in Sylhet and four cyclone landfalls on the coast, each window set from ReliefWeb situation reports (Table 4).

The masks were checked against an independent map of two of the same floods, the August 2017 flood in Rangpur and Rajshahi and the April 2017 haor flood in Sylhet, which the Global Flood Database maps from MODIS at 250 m (Tellman et al., 2021). On that grid, over cells that MODIS saw clear and that are not permanent water, the maps agree on 81.9 %

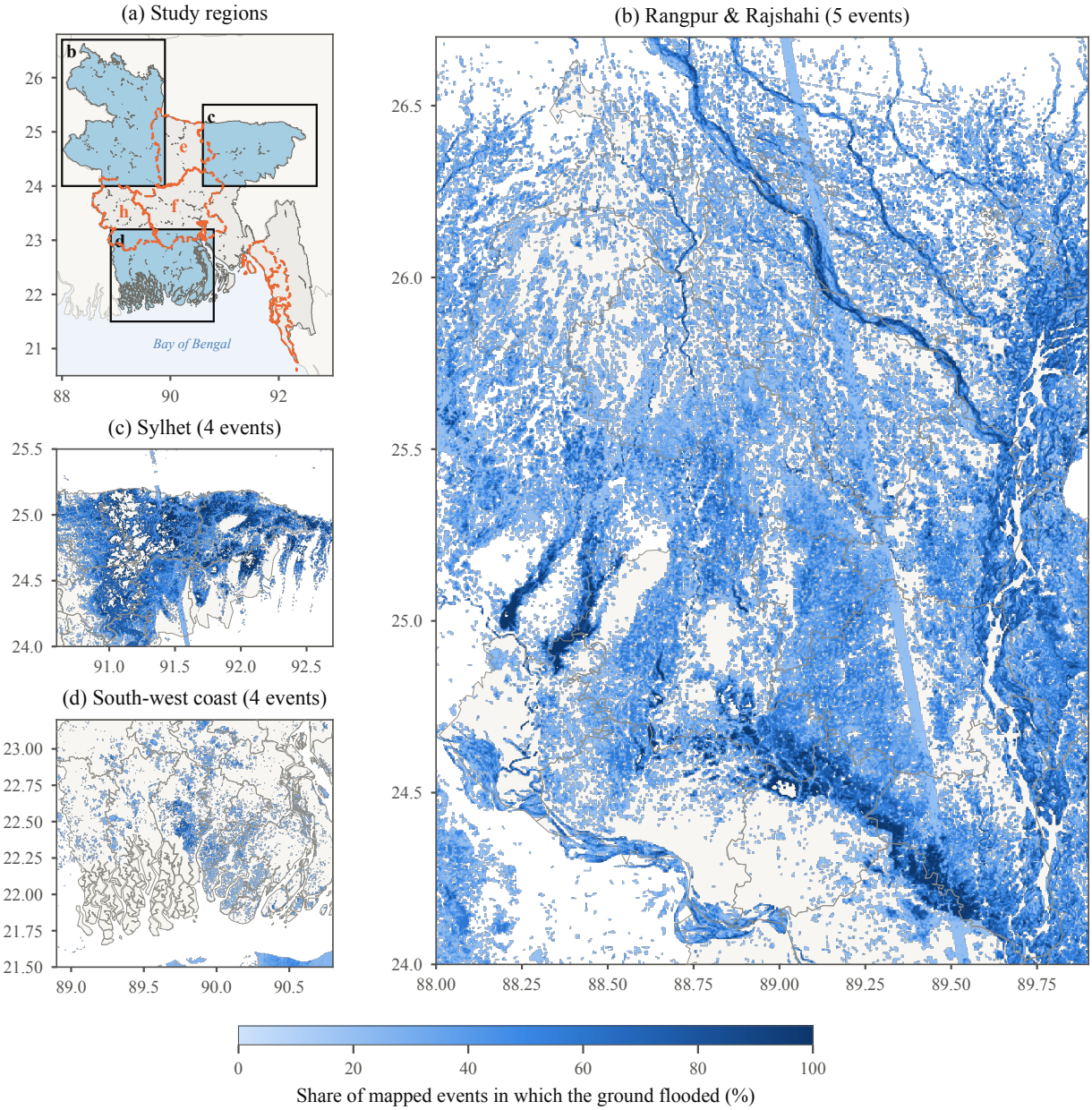


Figure 1. (a) The three study regions, drawn as their configured bounding boxes, and the validation region east of the Jamuna (e, dashed), over the district boundaries of geoBoundaries, with neighbouring countries from Natural Earth. (b–d) Share of the mapped Sentinel-1 events in which each 100 m cell was flooded. Cells never seen flooded are left blank, and the removal of permanent water leaves the main river channels empty. Straight-edged bands follow the edges of radar swaths.

Region	Hazard	Bounding box (W, S, E, N)	Configuration
Rangpur & Rajshahi	riverine flood	88.0, 24.0, 89.9, 26.7	`config.yaml`
Sylhet	flash flood	90.6, 24.0, 92.7, 25.5	`configs/sylhet.yaml`
South-west coast	coastal and tidal flooding	88.9, 21.5, 90.8, 23.2	`configs/sw_coastal.yaml`
Jamuna east[a]	riverine and flash flood	89.6, 23.9, 90.9, 25.5	`configs/validation/jamuna_east.yaml`
Central[a]	riverine flood	89.25, 22.8, 91.08, 24.39	`configs/validation/central.yaml`
Chattogram coast[a]	flash flood and surge	91.26, 20.54, 92.4, 23.04	`configs/validation/chattogram_coast.yaml`
West-central[a]	waterlogging	88.51, 22.75, 89.86, 24.24	`configs/validation/west_central.yaml`

Table 2. Configured study regions. [a]Validation region, defined by its districts within the box and without any asset a study box holds. Each box is restricted to Bangladesh, and all use UTM zone 46N (EPSG:32646) for distance and area calculations.

Dataset	Use	Resolution	Licence
OpenStreetMap	infrastructure assets (roads, bridges, schools, hospitals, shelters, embankments, cropland)	vector	ODbL 1.0
SRTM 1 arc-second	elevation, slope, topographic wetness, height above drainage, flow accumulation	30 m	public domain
JRC Global Surface Water	historical water occurrence, one proxy-label vote	30 m	free with attribution
WorldPop constrained 2020	population density for vulnerability and population exposure	100 m	CC BY 4.0
geoBoundaries gbOpen	districts (ADM2), upazilas (ADM3), unions (ADM4)	vector	CC BY 4.0
Sentinel-1 GRD (via Earth Engine)	observed flood extents for training labels and validation	10 m, exported at 100 m	Copernicus open

Table 3. Datasets ingested by the pipeline. National files (WorldPop, geoBoundaries, JRC tiles) are downloaded once into a shared cache and clipped per region.

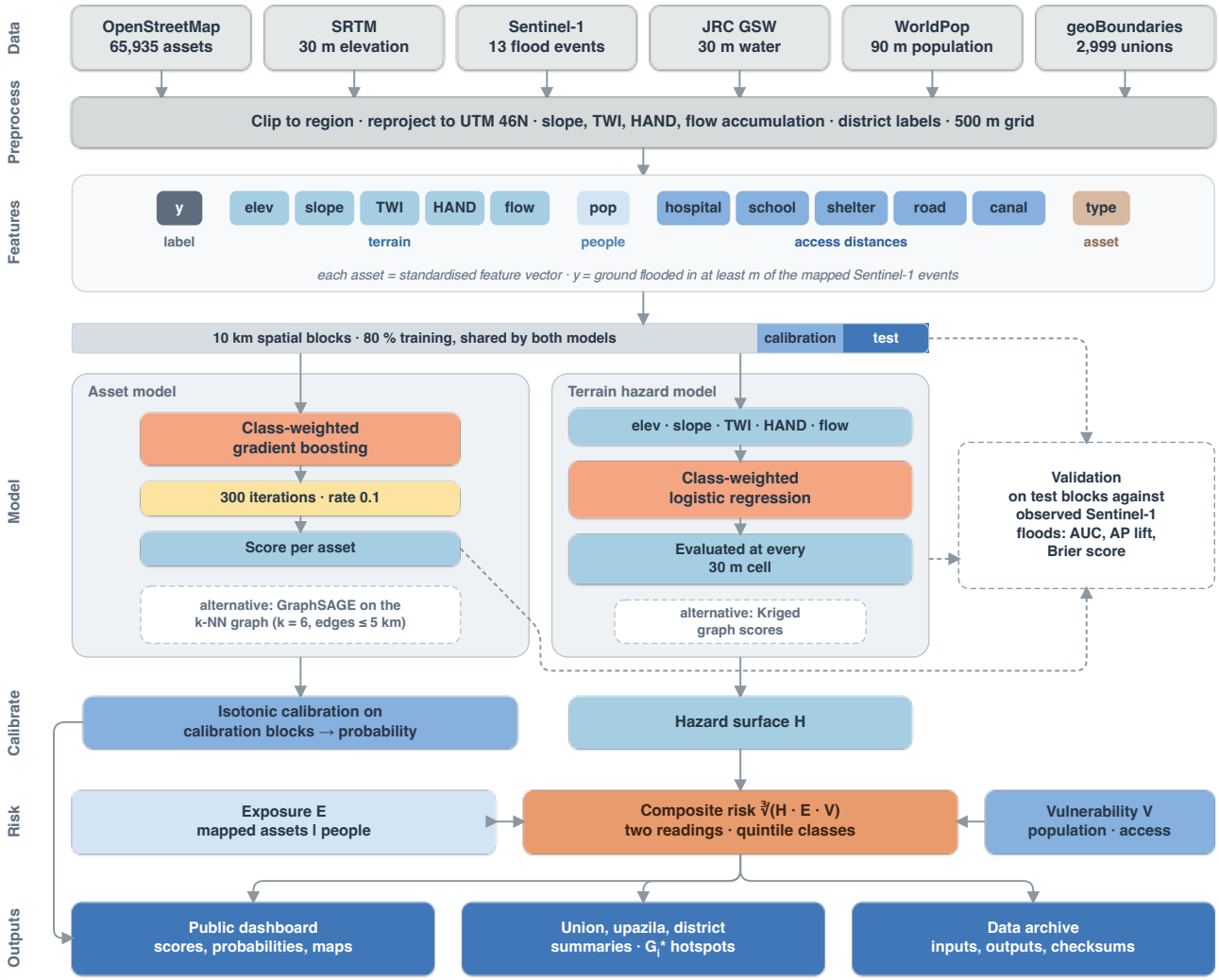


Figure 2. The processing chain, top to bottom. Source counts are totals over the three flood regions. Each asset enters the models as a vector of terrain, population, access-distance and asset-type features, with a label from the Sentinel-1 flood frequency. The asset model is expanded, with the graph network it replaced shown as the alternative it now is. Both models train on the same spatial blocks, and dashed connectors mark what the validation scores on the test blocks.

and 82.0 % of cells, a figure dominated by dry ground. On flooded ground the agreement is only moderate, with a critical success index of 0.290 and 0.339 and a kappa of 0.346 and 0.406. The radar marks more ground as flooded than MODIS in Rangpur and Rajshahi (19.7 % against 13.2 %), consistent with flooded patches smaller than a MODIS pixel and with flooding hidden by cloud. It marks less in the Sylhet haor (13.2 % against 23.3 %), consistent with the difficulty radar has where water stands under vegetation (Amitrano et al., 2024), as it did under unharvested rice in April 2017. Neither explanation was tested, and neither map is ground truth.

4.3 Training labels

In the riverine regions ground counts as flood-prone where it flooded in at least two of the mapped events, while on the coast one cyclone window is enough, because surge drains within days and the six-to-twelve-day revisit leaves too few pixels flooded twice. The pipeline also writes a terrain-threshold ensemble, which marks a pixel as flood-prone where at least two of three indicators agree, TWI above 8, HAND below 5 m and JRC water occurrence above 25 %. TWI and HAND are also model inputs, so a model trained and scored on these labels can succeed by relearning the threshold rule.

4.4 Asset model

The default asset model is class-weighted gradient boosting with 300 iterations and a learning rate of 0.1, which matched or beat a random forest, logistic regression and a two-layer GraphSAGE network (Hamilton et al., 2017) on a six-nearest-neighbour graph of assets, the design this system began with (Supporting Information, Section S7). Validation holds out whole 10 km blocks, assigned at random to training (80 %), calibration and test, because a random split leaves every test asset beside training assets with almost the same features and label. Scores are calibrated by isotonic regression on the calibration blocks, with each level shrunk towards the middle when it holds few points, and every metric comes from the test blocks. In the two riverine regions the published map scores come from the same model given one more feature, the share of earlier events in which the ground flooded (Section 5.3). Every validation figure outside that section comes from the model without it, because the feature is built from the events the models are scored against.

4.5 Validation design

Every model is scored on the held-out blocks by the area under the ROC curve (AUC), average precision and its lift over the base rate, and the Brier score. Every comparison between models or label sources is repeated over 20 block assignments drawn with different seeds. The paired differences are tested by the corrected resampled t-test of Nadeau and Bengio (2003), since the assignments reuse the same assets, and a plain t-test across them would overstate the evidence. The 10 km blocks are narrower than the ranges of the variograms fitted to the asset scores, roughly 115, 14 and 31 km, so test assets near a block edge still share terrain with training assets, and the main results are repeated with 20 and 40 km blocks on the same seeds (Section 5.5).

4.6 Later floods and published maps

The model and the record are compared on later floods in two designs. The first trains on the floods up to 2022 and scores against those of 2024 (Section 5.3). The second, a rolling origin, makes every flood year with an earlier event a test in turn. The model learns from the earlier years only, under the region's labelling rule capped by their number of events, the record is the share of those events in which the ground flooded, and both are scored on the held-out blocks against that year's floods. Each test also records the share of that year's flooded assets standing on ground no earlier flood had reached, termed new ground here, the quantity expected to decide which of the two ranks better. Four validation regions join the three study regions in this design, the Jamuna east bank, the central districts around Dhaka, the Chattogram coast and the west-central districts, each a region of a national partition of Bangladesh with every asset that a study box also holds removed, matched by OpenStreetMap identity or by location (Table 2). Distances there run to every facility of the region, including those removed.

Two published maps are scored on the same assets. The JRC global river flood hazard maps (Baugh et al., 2024), simulated with LISFLOOD-FP at about 90 m for return periods of 10 to 500 years, give each asset the annual exceedance probability of the most frequent flood that reaches it. JRC Global Surface Water occurrence (Pekel et al., 2016), the share of Landsat observations from 1984 to 2021 that saw water, stands for a record available anywhere without processing radar and is scored only against floods after 2021. Besides AUC, every ranking is scored by its Molchan trajectory and area skill score (Zechar and Jordan, 2008), with every asset weighted equally, so that a random ranking scores 0.5, and by the prediction accuracy index (Chainey et al., 2008) of a list as long as the record's, the share of flooded assets the list holds divided by the share of assets it flags.

5 Results

5.1 Held-out results

Table 5 reports one block assignment, the pipeline's default, which ranks 2, 1 and 17 of 20 by AUC in the three regions. In Sylhet 14.7 % of assets sit on flood-prone ground and the asset model reaches an AUC of 0.934. In Rangpur and Rajshahi only 6.7 % do, yet the model ranks well (AUC 0.884), with an average precision 7.0 times the base rate. Calibration lowers the Brier score from 0.069 to 0.029 in Rangpur and Rajshahi, slightly below the 0.032 of always predicting the base rate, and its deciles lie close to the diagonal in both riverine regions (Fig. 4). It carries less well to the coast, where the calibration blocks hold 5.5 % flood-prone assets against 1.7 % in the test blocks and the calibrated Brier score of 0.039 is worse than the base rate at 0.017.

5.2 Observed extents against terrain labels

The terrain-threshold ensemble captures 0.151 of the ground that flooded in at least 2 of the five Rangpur and Rajshahi events, with a false alarm ratio of 0.517, and marks 6.6 % of the region as flood-prone where the radar saw 21.2 % flood that often. Trained on those labels and scored on the same held-out blocks against observed flooding over 20 assignments, the default model reaches 0.716 against 0.817 for observed

Region	Window	Basis
Rangpur & Rajshahi	12–26 Aug 2017	second monsoon flood from 12 August; Dinajpur, Kurigram, Lalmonirhat and Rangpur among the worst affected
	12–26 Jul 2019	rain from 9 July, Brahmaputra–Jamuna rising to 18 July, Kurigram more than 1 m above danger level
	11–31 Jul 2020	second phase from 9–10 July, peak about 18 July, 27 basin gauges above danger into late July
	28 Jun–16 Jul 2024	northern floods from mid-June, Brahmaputra above danger at Nunkhawa, Hatia and Chilmari in early July
	27 Sep–12 Oct 2024	Teesta flooding of Lalmonirhat, Nilphamari, Kurigram and Rangpur from late September
Sylhet	30 Mar–20 Apr 2017	haor flooding from 28 March, striking Sunamganj on 1 April
	15–26 Jun 2022	heavy rain from 15 June, situation worst from 17 June
	16–28 Jun 2024	2.25 million affected in Sylhet, Sunamganj and Moulvibazar by 22 June
	18–30 Aug 2024	eastern floods from 20 August, including Moulvibazar and Habiganj
South-west coast	20–31 May 2020	Cyclone Amphan, landfall 20 May
	25 May–4 Jun 2021	Cyclone Yaas, landfall 26 May
	23 Oct–2 Nov 2022	Cyclone Sitrang, landfall 24–25 October
	26 May–5 Jun 2024	Cyclone Remal, landfall 26–27 May
Jamuna east	12–26 Aug 2017	second monsoon flood from 12 August
	13–28 Jul 2019	Brahmaputra–Jamuna rising to 18 July
	11–31 Jul 2020	second phase from 9–10 July, peak about 18 July
	3–15 Oct 2024	north-central flash floods from 3 October in Sherpur, Mymensingh and Jamalpur; first Sentinel-1 pass on 8 October
Central	15–31 Aug 2017	Faridpur, Rajbari, Manikganj and Munshiganj among the affected districts
	18 Jul–2 Aug 2019	monsoon flood reaching the central districts by 19 July
	15 Jul–10 Aug 2020	second phase of the 2020 monsoon flood
Chattogram coast	5–15 Aug 2023	flash floods and monsoon rain in Chattogram division
	26 May–1 Jun 2024	Cyclone Remal
	20 Aug–5 Sep 2024	eastern flash floods
West-central	13–30 Sep 2024	waterlogging of low-lying Jessore after heavy rain
	25 Jul–25 Aug 2025	waterlogging of the Bhabodah area

Table 4. Flood events mapped from Sentinel-1. Windows bracket the flood peak; the baseline for each is the February–March dry season of the same year (mid-January to March for the pre-monsoon haor event). Sources are ReliefWeb and United Nations situation reports and, for the west-central waterlogging, national newspapers, all cited in the configuration files.

Region	Assets	Positive rate	Asset AUC	Asset AP	Brier	Terrain AUC
Rangpur & Rajshahi	23,016	6.7 %	0.884	0.234	0.069	0.621
Sylhet	13,374	14.7 %	0.934	0.639	0.073	0.788
South-west coast	29,545	4.5 %	0.794	0.044	0.134	0.757

Table 5. Validation on held-out 10 km test blocks, for the pipeline's default block assignment, from the model without the past-flooding feature that the riverine maps carry (Section 4.4). Positive rate is the share of assets on ground the radar saw flood in the required number of events. The Brier score is that of the raw class-weighted scores, and the calibrated probabilities are reported in the text. Means over repeated assignments, lower in the riverine regions and higher on the coast, are in the Supporting Information, Table S3.

labels in Rangpur and Rajshahi, 0.770 against 0.857 in Sylhet and 0.758 against 0.839 on the coast. The gaps of 0.101, 0.087 and 0.080 favour the observed labels on 20, 19 and 19 of the assignments ($p < 0.001$, $p = 0.003$ and $p < 0.001$; Fig. 3). This test favours the observed labels, since that model is scored against the definition it learned, so Section 5.3 repeats it against later floods that neither label source was trained on.

5.3 Transfer to later floods

The held-out blocks separate training from testing in space but not in time. Each model is therefore also trained on the floods up to 2022, keeping each region's labelling rule capped by the number of earlier events, and scored on the held-out blocks against the ground any 2024 flood reached. Three references are scored on the same assets over 20 assignments (Table 6). The first is the share of earlier events in which the ground flooded, which is the map the radar record alone provides. The second is the same model trained with the 2024 events included, and the third is the same model trained on terrain-threshold labels, which draw on no flood event.

Trained only on earlier floods, the model ranks the 2024 flooding at 0.801 in Rangpur and Rajshahi, 0.797 in Sylhet and 0.807 on the coast. In both riverine regions the record of earlier flooding ranks it better, at 0.896 and 0.851, with the model ahead on only 2 and 4 of 20 assignments, whereas on the coast the model leads, at 0.807 against 0.668. The regions differ in how often floods return to the same ground, since only 16.9 % and 22.3 % of the assets flooded in 2024 stood on ground no earlier event had reached, against 65.5 % on the coast, where each surge floods the stretch of coast where it lands. The 2024 floods in Rangpur and Rajshahi were also small, covering 4.9–8.3 % of the region against 19.9–24.8 % for each earlier flood, whereas in Sylhet they were as large as earlier ones.

Seven checks test whether the comparison favours one side (Supporting Information, Section S5, Table S1), and in Rangpur and Rajshahi the record keeps its lead under all of them, if narrowly under two. Re-mapped from Sentinel-1A alone, since Sentinel-1B failed in December 2021 and fewer scenes can map less flooding, the earlier events shrank by up to 15.7 % and the record still led by 0.097. Retrained on flooding in any earlier event, which is how the 2024 floods are counted, the model fell further behind, by 0.133, and with 40 km blocks the lead grew to 0.126. In Sylhet the record's lead closed, to 0.017, when the model learned the same target as the test, so the Sylhet result rests on the choice of target, and on the coast the model's lead fell within noise with 20 and 40 km blocks.

East of the Jamuna, where the October 2024 flood reached only 43 assets, the record again led, at 0.881 against 0.713 on held-out blocks, but with a handful of flooded assets in most test blocks the lead fell short of significance ($p = 0.089$) except with 40 km blocks ($p = 0.044$). Out of fold over the whole region it led at 0.915 against 0.867, with a block-bootstrap interval for the difference from −0.129 to 0.019, and its list of 6.5 % of assets held 86.0 % of the flooded assets against 48.8 % for the model.

The record can also enter the model as a feature, provided it comes from floods before the ones the model learns to predict, or the model would copy its own labels. The model is therefore trained to predict the last event year up to 2022 from its usual features and the share of the earlier events in which the ground flooded, and scored against the 2024 floods with that share taken over every event to 2022. In Rangpur and Rajshahi the combination beats both of its parts, at 0.924 against 0.896 for the record alone ($p = 0.020$) and 0.790 for the model without the feature. Its gain comes on the ground no earlier flood reached, 87.7 % of the test assets, where the record gives every asset a score of zero and the combined model still ranks the 2024 flooding at 0.729. In Sylhet the combination, at 0.849, cannot be told apart from the model without the feature ($p = 0.196$) or the record alone ($p = 0.881$). On the coast, where the design leaves a single cyclone to train on, the combination at 0.648 cannot be told apart from the record alone ($p = 0.580$).

Measured as skill over the record (Supporting Information, Section S6), the share of the record's remaining ranking error that a model removes, the model scored −0.905 in Rangpur and Rajshahi, −0.364 in Sylhet and 0.416 on the coast, and 0.276 in Rangpur and Rajshahi given the record as a feature. A list of every asset an earlier flood reached held 93.2 % of the assets the 2024 floods reached in Rangpur and Rajshahi against 68.5 % for an equally long list from the model. On the coast, where the model ranked better overall, the record's list of 2.5 % of assets still held 37.1 % against 9.5 %, marking the most exposed ground more precisely even where it ranked the rest worse.

The 2024 floods also give the label comparison a test that favours neither label source. The observed labels still lead, at 0.801 against 0.732 in Rangpur and Rajshahi, 0.797 against 0.773 in Sylhet and 0.807 against 0.736 on the coast, ahead on 18, 15 and 18 of 20 assignments. The coastal gap of 0.071 stays close to its spatial 0.080, while the riverine gaps shrink, from 0.101 to 0.069 in Rangpur and Rajshahi and from 0.087 to 0.024 in Sylhet, where the remaining gap cannot be told apart from noise. Much of the Sylhet advantage in the spatial test therefore came from the shared definition rather than from the labels themselves.

5.4 Every flood year

Predicting every flood year from the years before it gives 15 tests, 8 in the study regions and 7 in the validation regions (Table 7). The record ranked the later flooding significantly better in 2, the model in 3, and in the other 10 the two could not be told apart, so the 2024 result in Rangpur and Rajshahi is not the rule for every year. The share of new ground decided which way a test went, ranging from 6.8 to 76.1 % (Fig. 5a). The record's lead fell as that share rose (Spearman $\rho = -0.775$, $p < 0.001$), in the study regions ($\rho = -0.833$) and the validation regions ($\rho = -0.679$, $p = 0.094$) alike, and the fitted line crosses zero at 33.4 % (95 % interval 18.2–41.1 %). New ground shrinks as the record lengthens, but the relation held with the number of earlier years held fixed (partial $\rho = -0.705$, $p = 0.003$), while that number added nothing once new ground was known ($\rho = 0.140$). In Rangpur and Rajshahi the model ranked the 2019 flood better, when 47.9 % of it reached new ground, the two were level in 2020, and the record led in 2024, at 6.8 %. Given the record as a feature, the model ranked the later flooding above both of its parts in 5 of 8 tests, all riverine.

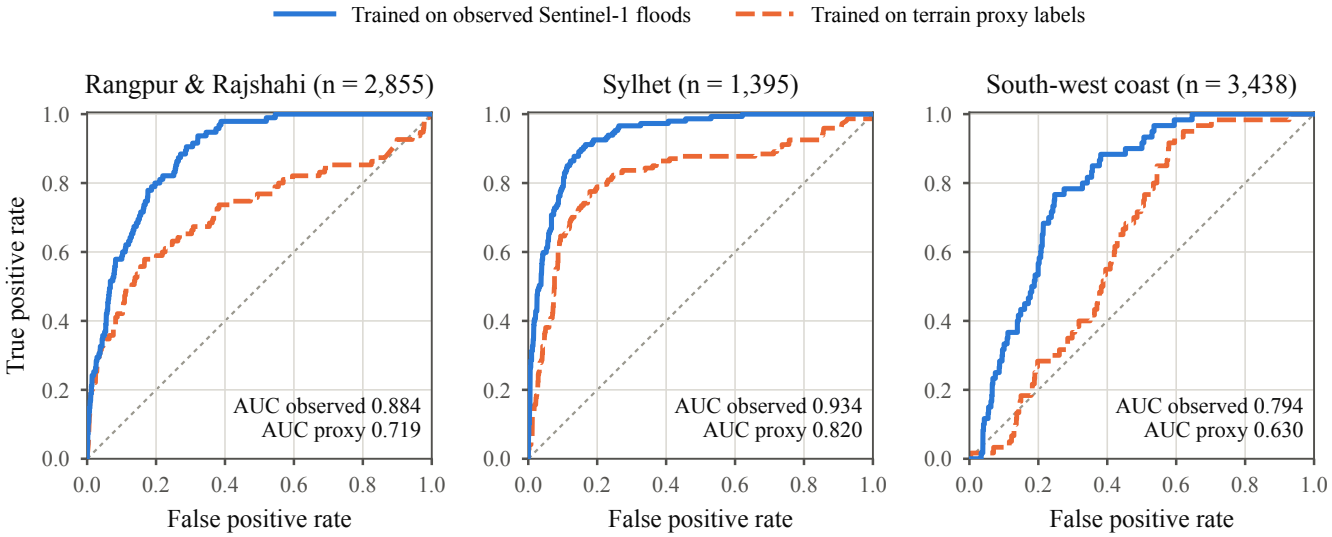


Figure 3. ROC curves on the held-out test blocks of each flood region, scored against the observed definition of flood-prone ground. Both curves come from the default gradient-boosted model on one block assignment, sharing features and blocks and differing only in their training labels. The dotted diagonal is a random ranking, and *n* is the number of assets in the test blocks.

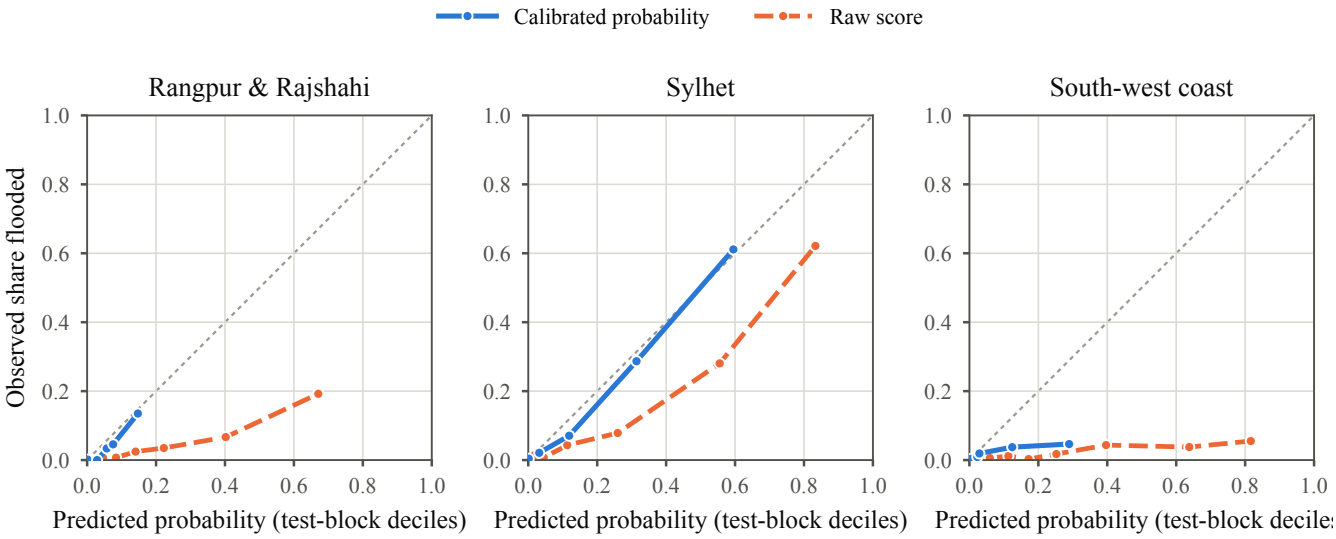


Figure 4. Reliability of the flood model on the test blocks, with assets grouped into deciles of each prediction. The raw class-weighted score overstates the likelihood of flooding. The isotonic calibration, fitted on other blocks, brings the riverine deciles close to the diagonal. On the coast, where the calibration blocks hold more flood-prone ground than the test blocks (5.5 % against 1.7 %), it still overstates the flooding observed.

Region	Model to 2022	All events	Past flooding	Terrain labels
Rangpur & Rajshahi	0.801 ± 0.063	0.794	0.896 ± 0.042	0.732 ± 0.067
Sylhet	0.797 ± 0.064	0.850	0.851 ± 0.039	0.773 ± 0.043
South-west coast	0.807 ± 0.055	0.834	0.668 ± 0.072	0.736 ± 0.066
Jamuna east (validation)	0.713 ± 0.243	0.720	0.881 ± 0.192	0.619 ± 0.184

Table 6. AUC against the 2024 flood extents on held-out 10 km test blocks, as mean ± standard deviation over 20 block assignments. The model to 2022 is trained only on the earlier events, the all-events model on labels that include 2024, past flooding is the share of the earlier events in which the ground flooded, and the terrain-label model is the same model trained on the terrain-threshold labels. The 2024 events reached 2.8 %, 17.7 %, 2.9 % and 1.1 % of assets in the four regions, and east of the Jamuna only 13 assignments left flooded assets in the test blocks.

The JRC hazard map ranked the later floods at 0.418 to 0.876, below chance on the coast, whose surges it does not simulate. It trailed the record significantly in 5 tests and led it in 4, each on a flood that reached much new ground, and the model ranked above it in 8 of 15. Landsat water occurrence, scored on the 8 floods after 2021, ranked them at 0.624 to 0.834, below the radar record in 4. Area skill scores ordered the record and the model as AUC did in 15 of 15 tests, and the Molchan trajectories show where the difference lay (Fig. 5b, c). In Rangpur and Rajshahi the record caught most of the 2024 flooding within the first sixth of the assets it flagged and then, holding nothing on the rest, ran down the diagonal, while on the coast the model missed fewer flooded assets at most list lengths.

5.5 Larger blocks

Wider blocks lower the held-out AUC of the default model only a little in the riverine regions, from 0.817 at 10 km to 0.791 at 40 km in Rangpur and Rajshahi and from 0.857 to 0.836 in Sylhet, and more on the coast, from 0.839 to 0.774, so the 10 km figures carry a small optimism, largest on the coast.

5.6 What the model relies on

Shuffling each feature across the test assets over the same assignments shows what the default model uses (Fig. 6). Elevation carries almost all of the terrain signal, costing 0.086, 0.220 and 0.129 in AUC when shuffled, while TWI, HAND, slope and flow accumulation cost at most 0.000, 0.009 and 0.007. Shuffled together, since correlated features can stand in for one another, the five terrain features cost 0.080, 0.214 and 0.132, about as much as elevation alone, and slightly less in Rangpur and Rajshahi, a difference within the noise of the shuffling. The five access distances together cost 0.110, 0.133 and 0.142, more than the terrain in Rangpur and Rajshahi and on the coast and second to it in Sylhet. They describe distance to services rather than any property of flooding and most likely stand in for where roads and settlements sit on the floodplain, a reading that was not tested directly.

6 Discussion

6.1 What the flood record already knows

Whether a susceptibility model anticipates a later flood better than the record of earlier ones depended, across 15 tests in seven regions, on how much of that flood reached new ground. Where less than about a third did, as in Rangpur and Rajshahi and Sylhet in 2024 and east of the Jamuna, the record ranked better or could not be told apart from the model, and where more did, as on the coast and in the first years of each record, the model ranked better in all but one test, significantly in 3. The share of new ground is known only after a flood, but in the riverine regions it fell as the record lengthened, from 47.9 to 6.8 % in Rangpur and Rajshahi and from 43.1 to 14.0 % east of the Jamuna, while on the coast, where each surge strikes its own stretch, it stayed near 62.9 %. It is therefore a property of the flood regime that a record of a few events already reveals rather than an accident of one year. Earthquake forecasting reached the same verdict for a hazard that also recurs where it struck before, with neither a pattern-recognition model nor a national hazard map gaining on past seismicity (Zechar and Jordan, 2008), and the hydrodynamic map scored here, which trailed the record significantly in 5 tests, shows that the comparison bears on hazard maps generally and not only on machine learning. A published multi-criteria map of Sylhet scored on the same 2024 floods reached 0.794 (Swarup and Nury, 2026), close to this model's 0.797, while the record ranked them at 0.851, a comparison that is only indicative, since that map was scored over areas.

The record marks where water returns and the terrain ranks where it has not yet been seen, and where floods recur the two combine well, the model given the record as a feature beating both in 5 of 8 tests. Floods returning to the same ground are expected on a floodplain, yet in Rangpur and Rajshahi a model scoring 0.817 on held-out blocks still ranked the 2024 flood below that expectation, at a skill over the record of −0.905. Forecasts are scored against persistence in the same way (Murphy, 1988; Pappenberger et al., 2015), and a susceptibility study that reports accuracy without that reference cannot show that its model knows more than the flood record already does.

6.2 Labels decide what an accuracy figure means

The label source decides what a published accuracy figure measures, since terrain-threshold labels cost the model 0.101, 0.087 and 0.080 in AUC on held-out blocks. The indices those labels are built on, TWI and HAND, add almost nothing to the model once elevation is known, and a model scored against labels derived from its own inputs measures its recovery of the labelling rule rather than its skill at locating floods. The mean AUCs here, 0.817 and 0.857 in the riverine regions, sit at or below the best-model AUCs of 0.87 to 0.98 in Table 1. The comparison is only indicative, since all but one of those studies split their points at random while every figure here comes from 10 km blocks the model never saw. Blocking has its own critics (Roberts et al., 2017; Wadoux et al., 2021), and the blocked scores estimate performance on unseen areas of similar terrain rather than design-based map accuracy.

6.3 Implications for flood risk management

First, where Sentinel-1 has observed several floods, the record of where water has already stood is a benchmark any susceptibility map should beat, and where floods return to the same ground it was the better guide to the next one. It costs little to compute, so agencies such as the Bangladesh Water Development Board and the Flood Forecasting and Warning Centre could publish it as a planning layer beside any susceptibility map. Where the record holds only one or two flood years, or where surges strike new stretches of coast, the model ranked better, and the two combined ranked best on the riverine floodplains, so the choice between them can follow how much of past floods fell on new ground. Second, a susceptibility map used to site shelters, embankments or schools should be judged on held-out areas and later floods rather than on random splits, since the accuracy a random split reports overstates how far the map can be trusted in places it has not seen.

6.4 Limits of the maps

The coast is the clearest limit, since the terrain hazard surface reaches only 0.610 there on held-out ground and the river hazard map ranks surge flooding below chance, because surge

Region	Year	Earlier years	New ground (%)	Record	Model	Model + record	Hazard map	Water occurrence	p
Rangpur & Rajshahi	2019	1	47.9	0.703	0.737	–	0.750	–	0.237
Rangpur & Rajshahi	2020	2	31.3	0.790	0.783	0.810	0.719	–	0.721
Rangpur & Rajshahi	2024	3	6.8	0.896	0.801	0.924	0.807	0.785	0.009
Sylhet	2022	1	76.1	0.601	0.696	–	0.729	0.708	0.009
Sylhet	2024	2	22.1	0.851	0.797	0.849	0.785	0.768	0.036
South-west coast	2021	1	30.8	0.788	0.822	–	0.434	–	0.710
South-west coast	2022	2	65.4	0.679	0.794	0.650	0.485	0.799	0.134
South-west coast	2024	3	62.9	0.668	0.807	0.648	0.418	0.624	<0.001
Jamuna east[a]	2019	1	43.1	0.739	0.735	–	0.876	–	0.955
Jamuna east[a]	2020	2	28.5	0.795	0.742	0.836	0.820	–	0.179
Jamuna east[a]	2024	3	14.0	0.881	0.713	0.920	0.872	0.834	0.089
Central[a]	2019	1	41.3	0.718	0.882	–	0.769	–	<0.001
Central[a]	2020	2	32.7	0.808	0.845	0.921	0.751	–	0.352
Chattogram coast[a]	2024	1	54.2	0.649	0.793	–	0.604	0.678	0.181
West-central[a]	2025	1	75.7	0.592	0.683	–	0.773	0.701	0.350

Table 7. Every flood year predicted from the years before it, as mean AUC on held-out 10 km blocks over 20 block assignments. New ground is the share of that year's flooded assets standing on ground no earlier flood had reached. The model given the record as a feature needs two earlier years, and the water occurrence ends in 2021, so it is scored only on later floods. The p-value is the corrected resampled t-test of model against record. [a]Validation region.

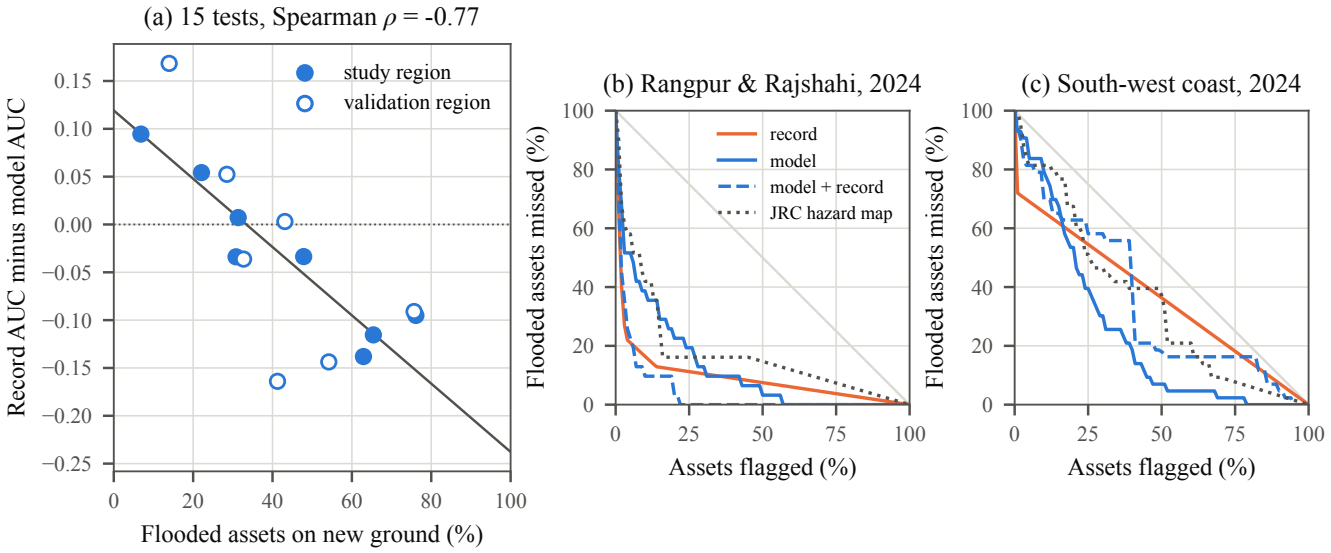


Figure 5. (a) The record's lead over the model in AUC against the share of each later flood that reached new ground, for every test in Table 7, with the least-squares line. (b, c) Molchan trajectories of the 2024 tests on one block assignment, the share of flooded assets missed against the share of assets flagged, lower being better; the grey diagonal is a random ranking.

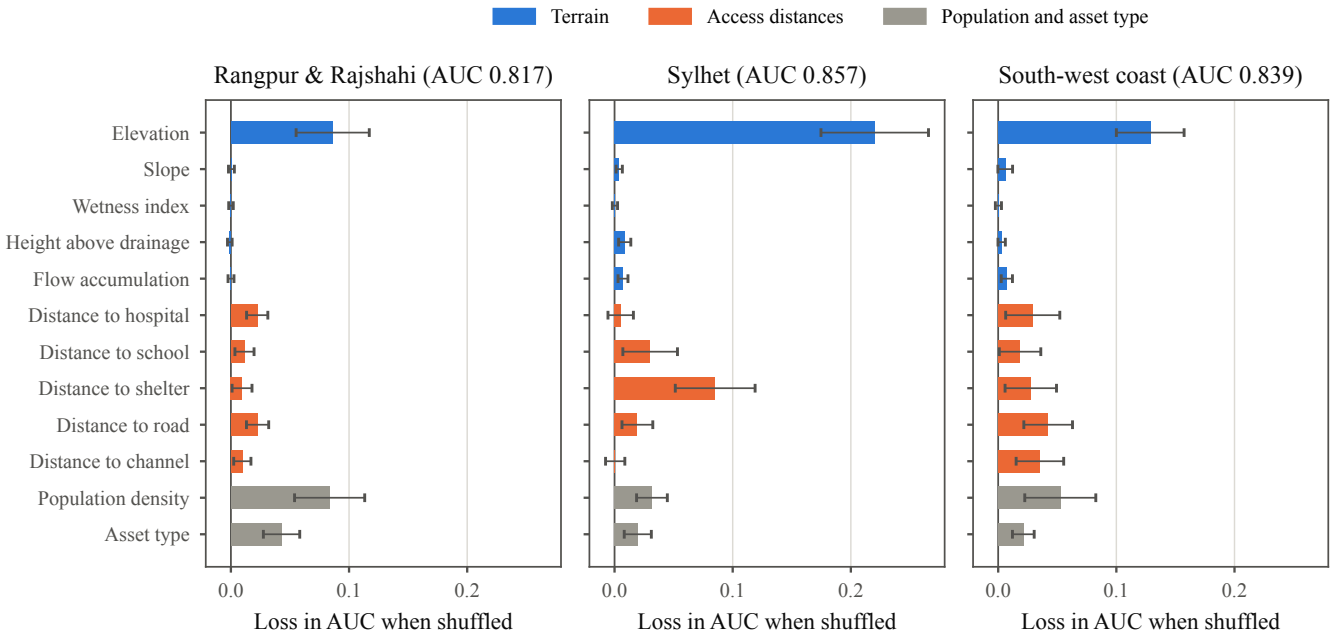


Figure 6. Loss in AUC on the held-out test blocks when each feature is shuffled across the test assets, as mean and standard deviation over 20 block assignments, for the default gradient-boosted model. The AUC in each title is the mean before shuffling.

follows embankments, tidal channels and landfall position, which neither describes. The composite risk index (Supporting Information, Section S2) depends on OpenStreetMap, whose rural completeness is unmeasured, and has not been checked against losses. The radar labels carry error of their own, agreeing only moderately with an independent MODIS map and counting inundated paddy as flooding, so every score here measures agreement with flooding mapped by radar rather than with flooding on the ground. Each test draws on one to three earlier flood years, and the threshold near a third of flooding on new ground is estimated from 15 tests, with an interval of 18.2 to 41.1 %. The calibrated probabilities hold for a region as a whole rather than for any one area, and none of the surfaces is a forecast.

7 Conclusions

A flood susceptibility model knew more than the flood record only where floods reached new ground. Across 15 tests in seven regions of Bangladesh, the record's lead over a terrain model fell as the share of each flood reaching new ground rose ($\rho = -0.775$), and the two were level at about 33.4 % of flooded assets. In Rangpur and Rajshahi the record predicted the 2024 flooding better than the model under every check applied, a global hydrodynamic hazard map was no stronger a rival, and a model given the record as a feature beat both of its parts in most riverine tests. Radar-observed flood extents made better training labels than terrain thresholds, by 0.101, 0.087 and 0.080 in AUC on held-out ground and by 0.069, 0.024 and 0.071 against later floods. Every number in this paper is generated from the released pipeline outputs, so each can be checked and rebuilt.

Data availability statement

The data that support the findings of this study are openly available in Zenodo at https://doi.org/10.5281/zenodo.23573843. The code that produced them is openly available at https://github.com/rakibhhridoy/rimes-mapathon under the MIT licence. The repository holds the pipeline, region configurations, dashboard and test suite. The pipeline also holds a landslide susceptibility module for the Chittagong Hill Tracts, which is described in the technical documentation and will be reported separately. The Zenodo archive packages the input data and outputs for the three study regions, the four validation regions and the Hill Tracts, per region with a manifest of checksums and licences, together with the rolling-origin outputs. Every figure in this paper is written into the text by the analysis scripts in the code repository directly from the pipeline outputs, so a rerun of the pipeline updates text and results together. The results are published through an open dashboard at https://fermium.systems/hazmapper, which shows the risk maps, rankings and per-unit summaries without a sign-in and states the limitations of Section 6 beside the maps. An earlier technical report on this system, prepared for the RIMES Mapathon in March 2026, contained errors that were found in a later review, the most serious being a coordinate-system error that left every training label at zero. Each error and its correction is listed in the technical documentation in the code repository, and no result of that report is used here.

Declarations

Competing interests. Md Rakib Hasan is affiliated with Fermium Systems, which hosts the public dashboard described here. The software is released under the MIT licence and the data under the open licences listed in Table 3, and no result depends on a proprietary component.

Author contributions. Md Rakib Hasan is the sole author and carried out the conceptualisation, methodology, software, formal analysis, investigation, validation, data curation, visualisation and writing.

Ethics approval and consent. Not applicable. The study uses public datasets only and involves no human participants.

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References

Adnan, M. S. G., Siam, Z. S., Kabir, I., Kabir, Z., Ahmed, M. R., Hassan, Q. K., Rahman, R. M., and Dewan, A. (2023). A novel framework for addressing uncertainties in machine learning-based geospatial approaches for flood prediction. *Journal of Environmental Management*, 326:116813.

Amitrano, D., Di Martino, G., Di Simone, A., and Imperatore, P. (2024). Flood Detection with SAR: A Review of Techniques and Datasets. *Remote Sensing*, 16(4):656.

Baugh, C., Colonese, J., D'Angelo, C., Dottori, F., Neal, J., Prudhomme, C., and Salamon, P. (2024). Modelled flood inundation for different return period scenarios at the global scale. European Commission, Joint Research Centre (JRC), dataset version 2.1.2.

Beven, K. J. and Kirkby, M. J. (1979). A physically based, variable contributing area model of basin hydrology / Un modèle à base physique de zone d'appel variable de l'hydrologie du bassin versant. *Hydrological Sciences Bulletin*, 24(1):43–69.

Chainey, S., Tompson, L., and Uhlig, S. (2008). The utility of hotspot mapping for predicting spatial patterns of crime. *Security Journal*, 21(1–2):4–28.

Chung, C.-J. F. and Fabbri, A. G. (2003). Validation of spatial prediction models for landslide hazard mapping. *Natural Hazards*, 30:451–472.

Dadhich, G., Babel, M., Miyazaki, H., Pal, I., Tripathi, N. K., and Virdis, S. G. P. (2026). Performance Evaluation of Automated Flood Inundation Mapping Techniques Using Multi-Temporal Sentinel-1 SAR Data. *Journal of Flood Risk Management*, 19(2):e70201.

Diptho, R. A., Ghosh, P., and Chowdhury, S. H. (2026). Physically consistent and spatially conformalized machine learning for trustworthy flood susceptibility mapping of the 2024 Feni flash flood. *Discover Geoscience*, 4:390.

Giezendanner, J., Mukherjee, R., Purri, M., Thomas, M., Mauerman, M., Islam, A. K. M. S., and Tellman, B. (2023). Inferring the past: a combined CNN–LSTM deep learning framework to fuse satellites for historical inundation mapping. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 2155–2165.

Guzzetti, F., Reichenbach, P., Ardizzone, F., Cardinali, M., and Galli, M. (2006). Estimating the quality of landslide susceptibility models. *Geomorphology*, 81:166–184.

Haghighi, M. H. (2022). Large-scale mapping of flood using Sentinel-1 radar remote sensing. In *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, volume XLIII-B3-2022, pages 1097–1102.

Hamilton, W. L., Ying, R., and Leskovec, J. (2017). Inductive Representation Learning on Large Graphs. Advances in Neural Information Processing Systems 30; arXiv:1706.02216.

Hawker, L., Neal, J., Tellman, B., Liang, J., Schumann, G., Doyle, C., Sullivan, J. A., Savage, J., and Tshimanga, R. (2020). Comparing earth observation and inundation models to map flood hazards. *Environmental Research Letters*, 15:124032.

Hawker, L., Uhe, P., Paulo, L., Sosa, J., Savage, J., Sampson, C., and Neal, J. (2022). A 30 m global map of elevation with forests and buildings removed. *Environmental Research Letters*, 17(2):024016.

Islam, A. R. M. T., Talukdar, S., Mahato, S., Kundu, S., Eibek, K. U., Pham, Q. B., Kuriqi, A., and Linh, N. T. T. (2021). Flood susceptibility modelling using advanced ensemble machine learning models. *Geoscience Frontiers*, 12(3):101075.

Islam, R. and Chowdhury, P. (2024). Local-scale flash flood susceptibility assessment in northeastern Bangladesh using machine learning algorithms. *Environmental Challenges*, 14:100833.

Jones, J. N., Boulton, S. J., Bennett, G. L., Stokes, M., and Whitworth, M. R. Z. (2021). Temporal variations in landslide distributions following extreme events: implications for landslide susceptibility modeling. *Journal of Geophysical Research: Earth Surface*, 126(7):e2021JF006067.

Murphy, A. H. (1988). Skill scores based on the mean square error and their relationships to the correlation coefficient. *Monthly Weather Review*, 116(12):2417–2424.

Nadeau, C. and Bengio, Y. (2003). Inference for the generalization error. *Machine Learning*, 52(3):239–281.

Niculescu-Mizil, A. and Caruana, R. (2005). Predicting good probabilities with supervised learning. In *Proceedings of the 22nd international conference on Machine learning - ICML '05*, pages 625–632.

Nobre, A., Cuartas, L., Hodnett, M., Rennó, C., Rodrigues, G., Silveira, A., Waterloo, M., and Saleska, S. (2011). Height Above the Nearest Drainage – a hydrologically relevant new terrain model. *Journal of Hydrology*, 404(1-2):13–29.

Pappenberger, F., Ramos, M. H., Cloke, H. L., Wetterhall, F., Alfieri, L., Bogner, K., Mueller, A., and Salamon, P. (2015). How do I know if my forecasts are better? Using

benchmarks in hydrological ensemble prediction. *Journal of Hydrology*, 522:697–713.

Pekel, J.-F., Cottam, A., Gorelick, N., and Belward, A. S. (2016). High-resolution mapping of global surface water and its long-term changes. *Nature*, 540(7633):418–422.

Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin, A., Gourlet-Fleury, S., and Pélissier, R. (2020). Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, 11(1):4540.

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., and Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929.

Rudra, R. R. and Sarkar, S. K. (2023). Artificial neural network for flood susceptibility mapping in Bangladesh. *Heliyon*, 9(6):e16459.

Sherpa, S. F. and Shirzaei, M. (2022). Country-wide flood exposure analysis using Sentinel-1 synthetic aperture radar data: Case study of 2019 Iran flood. *Journal of Flood Risk Management*, 15(1):e12770.

Swarup, S. D. and Nury, A. H. (2026). Flood risk prioritization in data-scarce regions: a tri-dimensional GIS-AHP framework for northeastern Bangladesh. *Journal of Water and Climate Change*, 17:1229–1256.

Talukdar, S., Ghose, B., Shahfahad, Salam, R., Mahato, S., Pham, Q. B., Linh, N. T. T., Costache, R., and Avand, M. (2020). Flood susceptibility modeling in Teesta River basin, Bangladesh using novel ensembles of bagging algorithms. *Stochastic Environmental Research and Risk Assessment*, 34(12):2277–2300.

Tellman, B., Sullivan, J. A., Kuhn, C., Kettner, A. J., Doyle, C. S., Brakenridge, G. R., Erickson, T. A., and Slayback, D. A. (2021). Satellite imaging reveals increased proportion of population exposed to floods. *Nature*, 596(7870):80–86.

Thomas, M., Tellman, E., Osgood, D., DeVries, B., Islam, A., Steckler, M., Goodman, M., and Billah, M. (2023). A framework to assess remote sensing algorithms for satellite-based flood index insurance. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16:2589–2604.

Twele, A., Cao, W., Plank, S., and Martinis, S. (2016). Sentinel-1-based flood mapping: a fully automated processing chain. *International Journal of Remote Sensing*, 37(13):2990–3004.

Uddin, K., Matin, M. A., and Meyer, F. J. (2019). Operational Flood Mapping Using Multi-Temporal Sentinel-1 SAR Images: A Case Study from Bangladesh. *Remote Sensing*, 11(13):1581.

UN-SPIDER (2019). Recommended Practice: Flood Mapping and Damage Assessment Using Sentinel-1 SAR Data in Google Earth Engine. https://un-spider.org/advisory-support/recommended-practices/recommended-practice-google-earth-engine-flood-mapping.

Wadoux, A. M.-C., Heuvelink, G. B., de Bruin, S., and Brus, D. J. (2021). Spatial cross-validation is not the right way to evaluate map accuracy. *Ecological Modelling*, 457:109692.

Zadrozny, B. and Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 694–699.

Zechar, J. D. and Jordan, T. H. (2008). Testing alarm-based earthquake predictions. *Geophysical Journal International*, 172(2):715–724.

Zzaman, R. U., Nowreen, S., Billah, M., and Islam, A. S. (2021). Flood hazard mapping of Sangu River basin in Bangladesh using multi-criteria analysis of hydro-geomorphological factors. *Journal of Flood Risk Management*, 14(3):e12715.